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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

IS2101-1 – COMPUTER ORGANIZATION & DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Floating point representation is used to store
 - A) Boolean Values
 - B) Whole Numbers
 - C) Real Integers
 - D) Integers
2. (2FA0C) 16 is equivalent to Binary is
 - A) (001011111010 0000 1100) 2
 - B) (001011111010 0000 1101) 2
 - C) (001011111010 0100 1100) 2
 - D) (001011111011 0000 1100) 2
3. Which of the following is NOT true about Normalization?
 - A) Floating point numbers are usually normalized
 - B) Exponent is adjusted so that leading bit (MSB) of mantissa is 1
 - C) Since it is always 1 there is no need to store it
 - D) Exponent is adjusted to 0 and divides it by 2
4. In the most Common and standard, device connected to the BUS are given addresses of _____ bit
 - A) 24
 - B) 64
 - C) 32
 - D) 16
5. The instruction Add LOCA, R0 does _____
 - A) Adds the value of LOCA to R0 and stores in the temp register
 - B) Adds the value of R0 to the address of LOCA
 - C) Adds the value of LOCA with a value in accumulator and stores it in R0
 - D) Adds the values of both LOCA and R0 and stores it in R0
6. The delays caused in the switching of the timing signals is due to
 - A) Memory access time
 - B) WMFC
 - C) Propagation delay
 - D) Processor delay
7. _____ are the different type/s of generating control signals.
 - A) Hardwired
 - B) Micro-instruction
 - C) Micro-programmed
 - D) Both Micro-programmed and Hardwired
8. Input or output devices that are connected to computer are called
 - A) Input/output Subsystem
 - B) Peripheral Devices
 - C) Interfaces
 - D) Interrupt
9. To overcome the lag in the operating speeds of the I/O device and the processor we use
 - A) Buffer Spaces
 - B) Status Flags
 - C) Interrupt Signals
 - D) Exceptions
10. _____ signal is sent by the initiator to indicate the duration of the transaction.
 - A) FRAME#
 - B) IRDY#
 - C) TMY#
 - D) SELD#
11. The classification of BUSES into synchronous and asynchronous is based on
 - A) The devices connected to them
 - B) The type of data transfer
 - C) The Timing of data transfers
 - D) Quality of Data Transfer
12. The process wherein the processor constantly checks the status flags is called as
 - A) Polling
 - B) Inspection
 - C) Reviewing
 - D) Echoing

13. The device which starts data transfer is called .

- A) Slave
C) Distributor
B) Transactor
D) Master

14. The number successful accesses to memory stated as a fraction is called as

- A) Access rate
C) Hit rate
B) Success rate
D) Miss rate

15. The return address from the interrupt-service routine is stored on the _____

- A) System heap
C) Processor stack
B) Processor register
D) Memory

16. The smallest entity of memory is called

- A) Cell
C) Instance
B) Block
D) Unit

17. To get the physical address from the logical address generated by CPU we use

- A) MAR
C) Overlays
B) MMU
D) TLB

18. CMOS cell is used for _____ memory

- A) static memory
C) Cache memory
B) dynamic memory
D) Ram memory

19. _____ and _____ are two classifications of RAM memory

- A) static memory and dynamic memory
C) Cache and Flash memory
B) ROM and EPROM
D) Virtual memory and Real memory

20. _____ method is used to map logical addresses of variable length onto physical memory.

- A) Paging
C) Segmentation
B) Overlays
D) Paging with segmentation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) With a neat diagram name and explain the Functional Units (FU). Also explain the conditional status registers.	8	L*2	1	1
b) Write the Booth's algorithm and explain the booth algorithm with -13 x 11 as an example	8	L2	2	1
2. a) Explain the operational concepts while showing the connection between processor register and memory? Give an example and explain with the steps, how the data transfer happens in between the processor and memory.	8	L2	1	1,2
b) Demonstrate the working of the Restoring Division algorithm. Apply the same on 15/4.	8	L2	2	1
3. a) Explain how the performance of the processor is calculated? Give a formula for the performance evaluation? And how the changing of parameter in the formula affects the performance of the system? How the SPARC performance is evaluated? A computer system has a CPU with a clock rate of 5 GHz and an instruction mix with the following CPI values: ☐ 20% of the instructions have CPI = 1 ☐ 40% of the instructions have CPI = 2 ☐ 20% of the instructions have CPI = 3 ☐ 20% of the instructions have CPI = 4 Determine the average CPI of the system and the CPU execution time per instruction.	8	L2	1	1

- b) i. Write a short note on IEEE format for Floating point representation. Apply IEEE single precision floating point notation on 100.25.
 ii. What is the difference between the restoring and non-restoring division?

8 L2 2 1

Unit – II

4. a) Explain with block diagram the hardwired control unit organization using separate decoding and encoding functions.
 b) What is an interrupt? Explain how interrupt is handled by the processor? Also explain the enabling and disabling of the interrupt?

8 L2 3 1

8 L2 4 1

5. a) With neat diagram, explain the three bus organization and control steps involved in complete execution of ADD R3, R4, and R5.
 b) What is bus arbitration? Explain different types of bus arbitration with a neat diagram.

8 L2 3 1

8 L2 4 1

6. a) Explain the basic organization of micro programmed control unit? Write a micro routine for the instruction BRANCH<0?

8 L2 3 1

- b) Write a short note on

i. USB

ii. PCI

8 L2 4 1

Unit – III

7. a) Differentiate between ROM and RAM? Explain different types of ROM. Write the memory Hierarchy diagram with mentioning the speed and size of the memory and explain it.

08 L2 5 1

- b) Why caching is required? What is cache hit and miss? Explain different performance consideration methods used to improve the performance of the cache.

08 L2 5 1

8. a) Why replacement algorithms are required? Explain different types of Cache replacement algorithms with example

08 L2 5 1

- b) What is virtual memory? Why it is required? With a neat diagram explain the virtual memory organization.

08 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
